

VisionTrust Assurance

Offline Computer-Vision Integrity & Provenance Framework

Report ID: REP-C2F09027B476 | Generated: 2026-09-22T08:55:13.998346+00:00 (Air-Gapped)

Overall Assurance Verdict

ACCEPT

Overall Risk Score: 0.0 / 100 (LOW)

Executive Summary: No integrity violations or anomalies detected.

Metric	Value	Metric	Value
Critical Findings	0	Dataset Sample Count	49
High Findings	0	Dataset Classes	3
Medium Findings	0	Model Access Mode	N/A
Low Findings	0	Model Parameter Count	N/A
Inference Records Tracked	1	Tampered Inferences	0

Integrity & Security Findings

No critical findings or integrity violations reported.

Assurance Threat Coverage Statement

ID	Attack / Threat Capability	Status	Access Mode	Primary Method
T1	Label Flipping Detection	SUPPORTED	Dataset Labels + I	Feature k-NN neighborhood consensus & imbalan...
T2	Systematic Mislabelling	SUPPORTED	Dataset Metadata +	Hierarchical contributor & batch anomaly roll...
T3	Duplicate Flooding	SUPPORTED	Image Files (Black	Perceptual Hashing (pHash, dHash) Hamming dis...
T4	Out-of-Distribution (OOD) Injection	SUPPORTED	Dataset or Referen	Isolation Forest & statistical feature distri...
T5	Trigger / Backdoor Injection	PARTIALLY_SUPPORTED	Dataset Images	Localized patch spatial consistency & class c...
T6	Model Substitution	SUPPORTED	Model File (.onnx,	Cryptographic SHA-256 identity & parameter co...
T7	Model Weight Modification	SUPPORTED	Model Weights (Whi	Tensor statistics, layer shapes, and mean L2-...
T8	Backdoor Behaviour Testing	SUPPORTED	Inference Executio	Controlled perturbation battery (brightness, ...
T9	Inference Record Tampering	SUPPORTED	Full Record Payloa	Cryptographic hash binding + Ed25519 digital ...
T10	Inference Replay Attacks	SUPPORTED	Inference Log Stat	Cryptographic nonces, monotonic sequence chec...
T11	Provenance Configuration Substitution	SUPPORTED	Provenance Metadat	Canonical JSON serialization hashing for prep...
T12	Distribution Shift & Drift	SUPPORTED	Reference & Candid	PSI, 2-sample KS test, Wasserstein distance, ...

System Limitations & Air-Gap Compliance

- Framework runs 100% locally and does not transmit data or consult cloud foundation models.
- Statistical heuristics cannot mathematically prove total absence of steganographic or adversarial latent triggers.

- Model weight analysis on ONNX models depends on initializers embedded within the graph file.
- Offline cryptographic protection guarantees tamper-evident provenance but assumes private key is stored securely in air-gapped host.